

3. Zoho

Round 1 - Selected

General Apti + C Snippets

ZOH0 (Round 1 - writtentest - 2024)

- 1) If 2 fair coins are flipped atleast 1 of outcomes is known to be a head, what is prob. of both are heads
- 2) $f(x+y) = f(x) + f(y)$ for all +ve integers of x & y , If $f(2) = 4$, what is $f(3)$?
- 3) A tortoise at bottom of 65 m hill is trying to climb. For every hour, it climbs 5 meters & slips down 3 m. How long will it take to climb.
- 4) If $7^x - 7^{x-2} = 49(7^6)$. Find x
- 5) A 3 digit no. is sum of unit digits & hundreds digit is = to tens digit. If sum of digits is 12 what is the longest number if none of digits are zero
- 6) dia of circle is increased by 50%, what % does area ↑.
- 7) In how many diff way 3 identical green shirts & 3 identical red shirts can be distributed among 6 children
- 8) $\sqrt{x+6} + \sqrt{x+1} = 5$ what is x^2
- 9) If a group of students having an average of 16 yrs joined a class, the avg. of age of the students in the class reduces from 18 to 17. What is the ratio of the no. of students who joined the class to the no. of students who were initially in class?
- 10) $\sqrt{\sqrt{3x}} = 4\sqrt{2x}$

Round 2 - Selected

1. Rotate a matrix 180 Degree
- 2.

6. Input: n =
5 Output:

1	3	6	10	15
2	5	9	14	19
4	8	13	18	22
7	12	17	21	24
11	16	20	23	25

3. Find the longest palindromic substring.

IP -> LIRILSOAP

OP -> LIRIL

4. Represent the given number in terms of minimum possible pseudo binary numbers

IP - 32

OP - 11 + 11 + 10

5. Find the shortest possible path of a king to travel from its initial position to its target position in a chess board.

IP :

Source : { 1, 1 }

Target : { 2, 4 }

OP :

(1,1) -> (2,2) -> (2,3) -> (2,4)

L2 Questions at Incubation :

1.

1	2	3	4	5	6	7
14	13	12	11	10	9	8
15	16	17	18	19	20	21
28	27	26	25	24	23	22
29	30	31	32	33	34	35
42	41	40	39	38	37	36
43	44	45	46	47	48	49

2.

					1							
					2	1	2					
				3	2	1	2	3				
			4	3	2	1	2	3	4			
		5	4	3	2	1	2	3	4	5		
	6	5	4	3	2	1	2	3	4	5	6	
7	6	5	4	3	2	1	2	3	4	5	6	7

3. Given two sorted (ascending or descending) arrays, We have to merge those arrays in the ascending order

4. Find the number till it becomes single digit

Ex : 45

$4 + 5$

9

Ex :

32

$3+2$

5

Ex :

79

$7+9 = 16$

$1+6 = 7$

5. Sort a array based on its factors

6. Find the next smallest palindrome of a given number

Ex 1 : 99

OP : 101

Ex 2 : 1721

OP : 1771

7. Rearrange the given array such that first maximum at first position, second maximum at last position , third maximum at second and so on..

IP : [4, 6, 1, 2, 5]

OP : [6, 4, 1, 2, 5]

8. Arrange given numbers to form the biggest number

Input: $n = 5$, $arr[] = \{3, 30, 34, 5, 9\}$

Output: 9534330

9. Langton's ant

10. Screensaver Based Question

L3 Questions :

1.

Given family and extended family information, I was asked to construct a family tree from that information and need to find the list of eligible people one can marry. The conditions were as follows:

For a male, a match will be his father's sister's daughter or his mother's brothers' daughter.

For a female, a match will be her father's sister's son or her mother's brother's son. Names are unique.

Input format:

<person's name>, <gender>, <father's name>, <mother's name>

Input:

John, Male, Brad, Lisa

Emma, Female, Brad, Lisa

Alex, Male, John, Jenny

Emily, Female, Steve, Emma

Rachel, Female, Steve, Emma

Person name: Alex

Output: Emily, Rachel

2. Multiply the given equations

$$(2x + 3y)(x + y^2)$$

Answer must be multiplication of above equations

3. Minesweeper Game

All L3 questions must be designed with OOPS

Round 4 and 5 :

1. Self Introduction
2. Will you opt for higher studies
3. What if you get rejected
4. Salary Expectations
5. Aim for Next 10 years
6. What do you expect from Zoho
7. Asked about family details